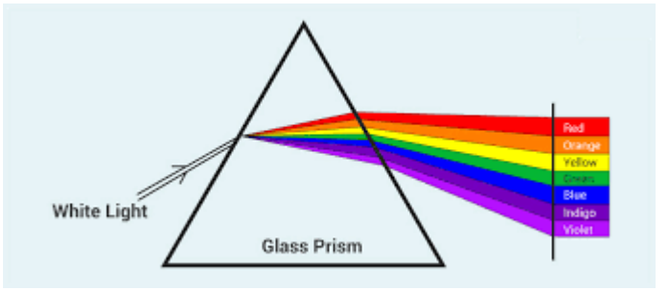
Knowledge Organiser
Year 8: Waves – Light Waves

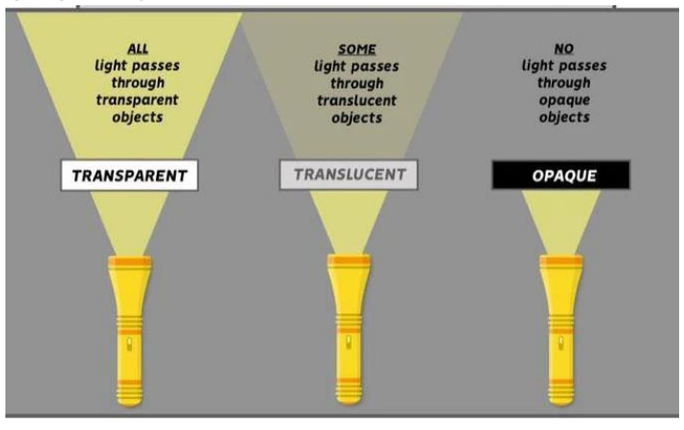
The law of reflection… **Angle of incidence** = **Angle of reflection**

How does light behave?
- Light travels in straight lines as a transverse waves
- Light is part of the electromagnetic spectrum, all EM waves travel at 300,000,000 m/s
- White light is made of a mixture of all the different colours found in visible light

A diagram showing how a glass prism can split up white light into all the different wavelengths



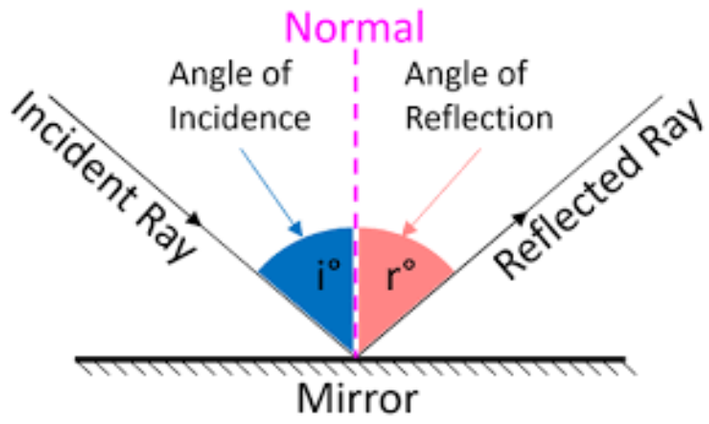
A diagram comparing transparent, translucent and opaque objects



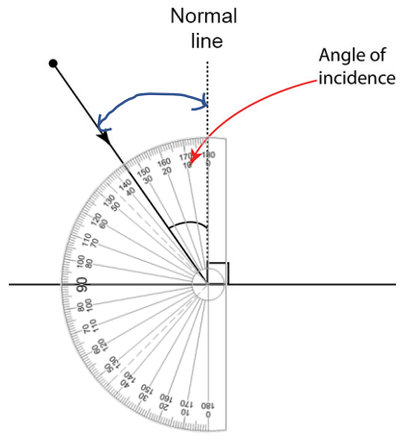
How does reflection work?
- Light waves can be reflected, which means to throw back without absorbing
- The rays of light follow the law of reflection when they hit a reflective surface

What is the law of reflection?
- The law of reflection states that the angle of incidence is equal to the angle of reflection
- These angles are measure from the **normal line**
- This is a line which is drawn **perpendicular** (at a right angle to) **the** position that the light ray hits the reflective surface

A diagram which illustrates the law of reflection



A diagram which shows how to measure the angle of incidence

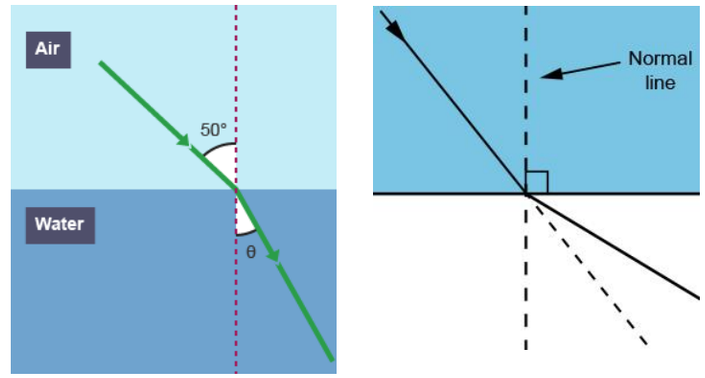


How to measure the angles:
- Draw a normal line at 90 degrees from the mirror plane
- Place your protractor lined up with the normal line
- Measure the angle from 0 to the ray (shown in blue)

What is refraction?
- When a ray of light travels from one material to another it can change its speed
- When the ray changes speed it will also change direction, this is called **refraction**
- The direction and size of the change depends on the difference in **refractive index** between the two materials

Drawing ray diagrams for refraction:
- If the refractive index of the new material is higher, the ray will bend towards the normal
- If the refractive index of the new material is lower, the ray will bend away from the normal

Two ray diagrams showing a ray moving from 1. a low to high refractive index and 2. a high to low refractive index

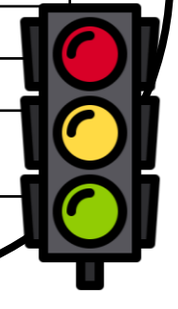


How do we see coloured objects?
- When light hits an object, only one frequency is reflected, all others are absorbed by the object
- If no light is reflected by the object, it will appear black
- If all frequencies of light are reflected by the object, it will appear white



RAG Yourself!

Learning Outcomes;	RAG
Compare the speed of sound and light	
Define the key terms transparent and opaque	
State the law of reflection	
Draw ray diagrams to illustrate the law of reflection	
Measure the angle of incidence on a ray diagram	
Explain how coloured objects are seen	
Explain why light is refracted when it travels between mediums	



Knowledge Organiser
Year 8: Waves – Light Waves

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The law of reflection…

What is the law of reflection?

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Drawing ray diagrams for refraction:

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How to measure the angles:

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What is refraction?

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